

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2020

Bachelor in Computer Applications

Course Title: **Basic Statistics**

Code No: Semester: **2**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. What is the difference between relative and absolute statistical measure? J.J. Thomson discovered the electron by isolating negatively charged particles for which he could measure the mass-charge ratio. This ratio appeared to be Constant over a wide range of experimental conditions and, consequently, could be characteristics of new particles. His observations; from two different Cathode- ray tubes that used air as the gas. [10]
Tube I 0.570.340.430.320.480.400.40
Tube II 0.530.470.470.510.630.610.48
Which tube is more homogeneous?
2. In the Nepal, during the fall harvest season Pumpkin are sold in the large quantities at the farm stands. Often instead of weighting the Pumpkin prior to sale, the farm stand operators will just places the Pumpkin in the appropriate circular cutout on the counter. When asked why this was done one farmer replied. can tell the weight of the Pumpkin from its circumference". To determine whether this was really true; a sample of 8 Pumpkin's circumference and weights were measured: [10]
Circumference (cm) 50.55545237525347
Weight (gram) '00'1220151750101514a. Develop the estimating linear regression equation of weight of the Pumpkin to the circumference, and b. Estimate the weight of Pumpkin if the circumference value is 40 cm. Can it be possible to predict the weight of a pumpkin if it's circumference if 80 cm?
3. Define normal distribution and standard normal distribution. In an examination, 10% students got less than 20 marks and 95% got less than 75 marks. Assuming the distribution to be normal, find the mean and standard deviation. [10]
4. What do you mean by secondary data? Write the source of secondary data with special focus to Nepalese context. [5]
5. Construct histogram, frequency polygon and frequency curve of the following frequency distribution. [5]
Marks 0-20 20-40 40-60 60-80 80-100
No. of Students 10 22 35 28 5
6. Following data represent the preference of 10 students studying BIT towards two brands of computers namely DELL and HP. [5]
Computer Student preference
DELL 52981103467
HP 10513862794
Apply Pearson's rank correlation coefficient and compute correlation coefficient. Also interpret your result.
7. Define random variable. A random variable has the following probability function. [5]
X -2 -1 0 1 2
P(X) 0.2 0.4 0.2 0.1 0.1
Find (i) K, (ii) P (X ≥ 0), (iii) E(X), and (iv) E (4X+5).

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8. During one stage in the manufacture of integrated circuit chips, a coating must be applied. If 70% of chips received a thick enough coating, find the probability that among 15 chips (i) at least 12 will have thick enough coatings, (ii) at most 2 will have thick enough coatings, and (iii) exactly 10 will have thick enough coatings. [5]
 9. A new computer virus can enter the system through e-mail or through the internet. There is a 30% chance of receiving this virus through e-mail. There is a 40% chance of receiving it through the internet. Also, the virus enters the system simultaneously through e-mail and the internet with probability 0.15. What is, the probability that the virus does not enter the system at all? [5]
 10. The standard deviation of a symmetric distribution is 9. Compute the possible value of fourth central moment for the distribution to be (i) mesokurtic (ii) platykurtic, and (iii) leptokurtic. [5]
 11. The mass of vitamin E in a capsule manufactured by a certain drug company is normally distributed with standard deviation 0.042 mg. A random sample of 36 capsules was analyzed and the mean mass of vitamin E was found to be 5.12 mg. Calculate a symmetric 95% confidence interval for the population mean mass of vitamin E per capsule. [5]
 12. Write short notes on the following: [2.5x2=5] a. Box and whisker plot. b. Census survey and sample survey.